IT-arkitektur och IT Service Management

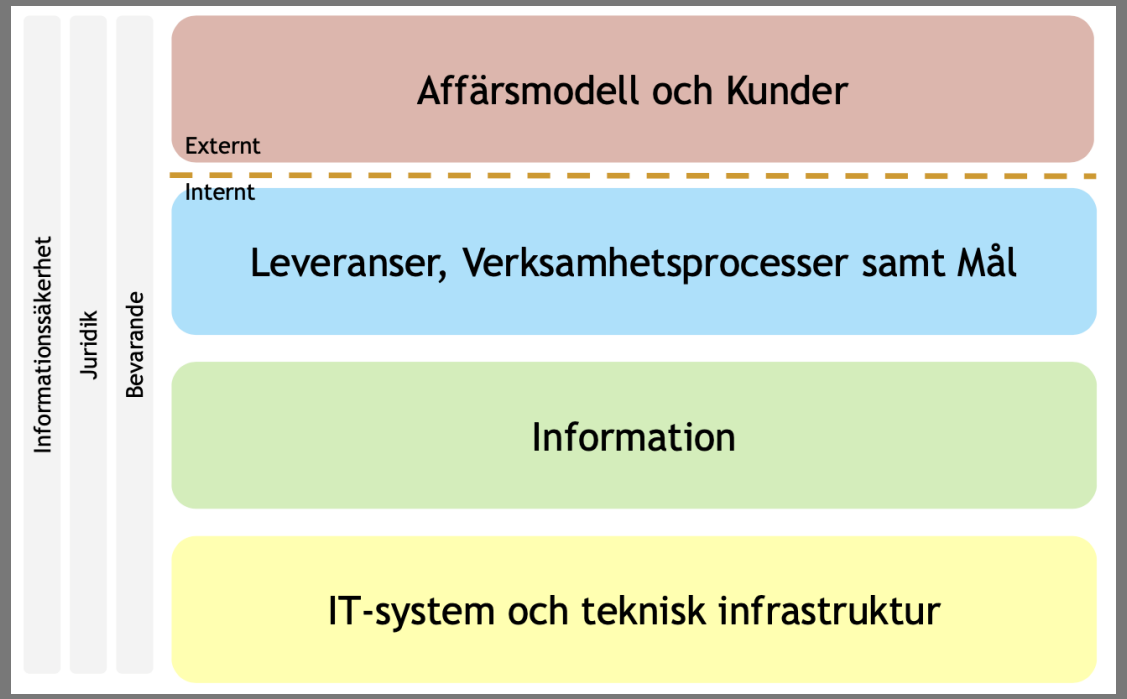
Modellering i ArchiMate

Innan vi börjar…

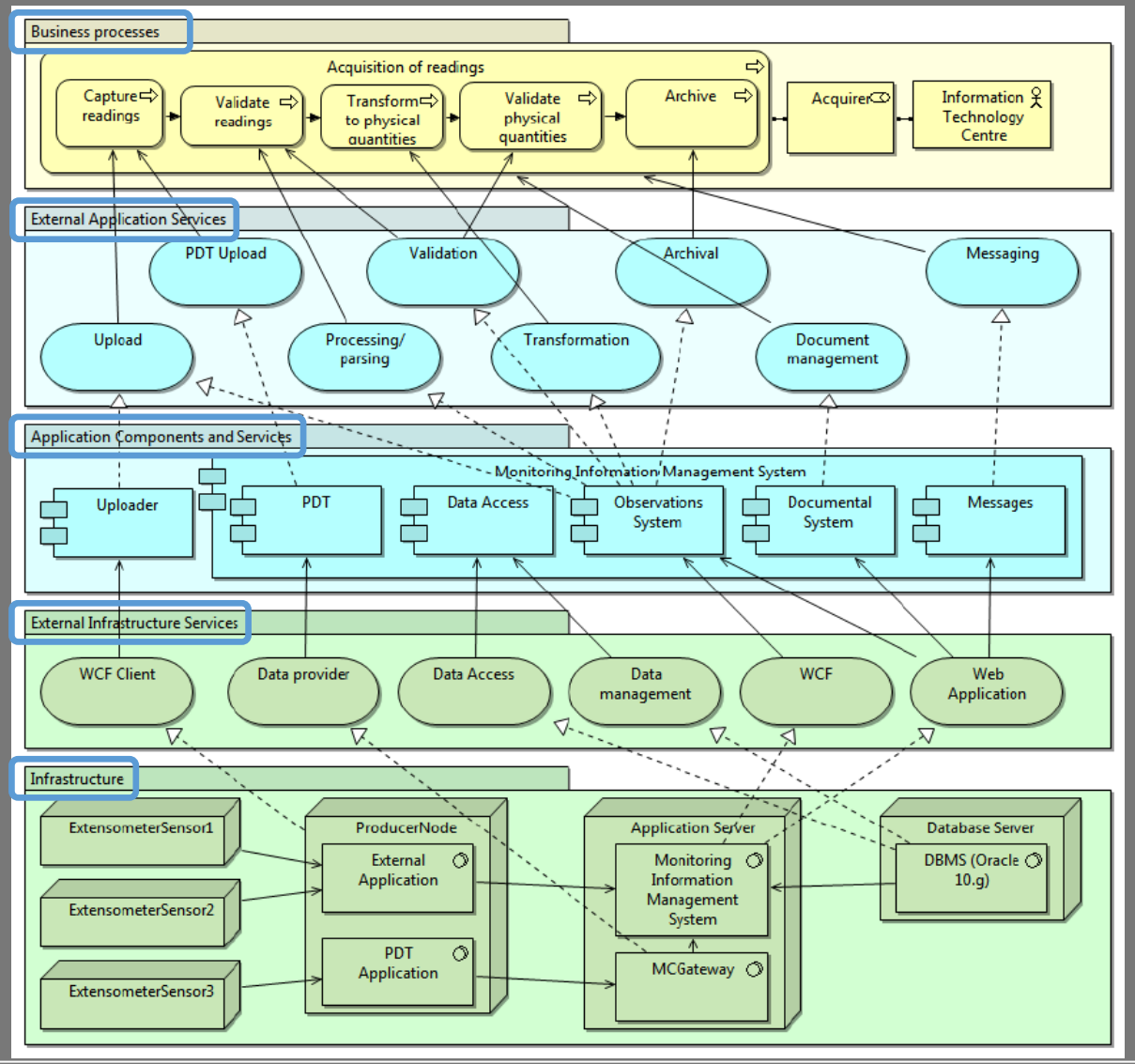
EA CONSTITUENTS

- **EA in different layers**
  - Business
  - Information
  - Applications
  - Technology

  } Information systems

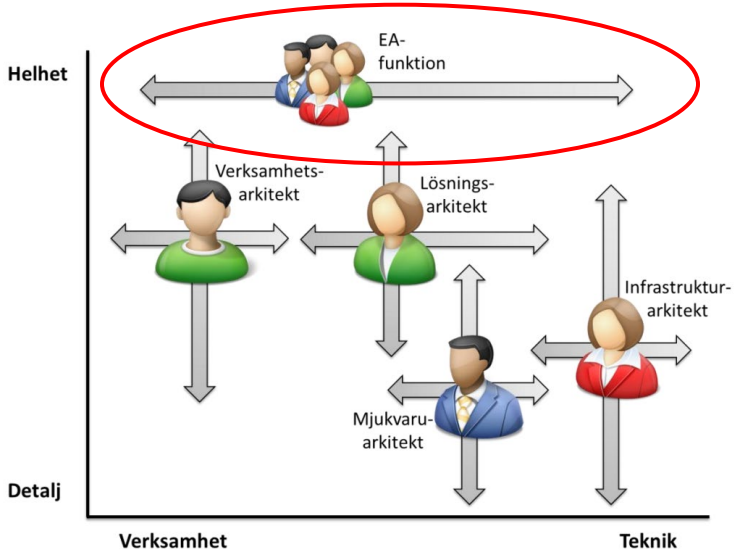


EX1 - ENTERPRISE ARCHITECTURE MODEL



Alignment

EA-funktionen

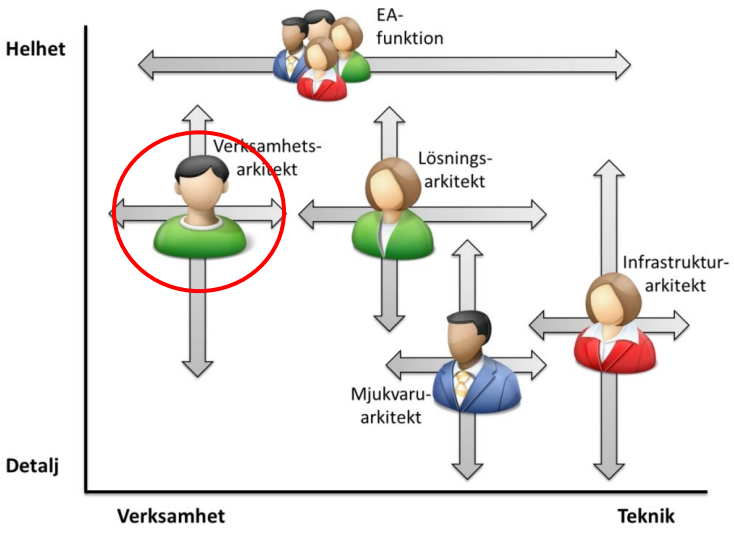


- Sammanhållande och övergripande funktion för allt arkitekturarbete inom organisationen
- Ibland en specifik roll, men oftast:
  - En funktion som uppstår när samspelet mellan olika arkitektroller fungerar väl
- Säkerställer att IT på totalen stödjer organisationens syfte, mål och strategier – hög effekt och maximal nytta av investeringar inom IT

Enterprisearkitektens output

- Den samlade modellen/arkitekturen för en organisation och de informationssystem om används (EA-modell)
- En vanlig leverabel är **beslutsunderlag** och **styrmodeller** för ledningen

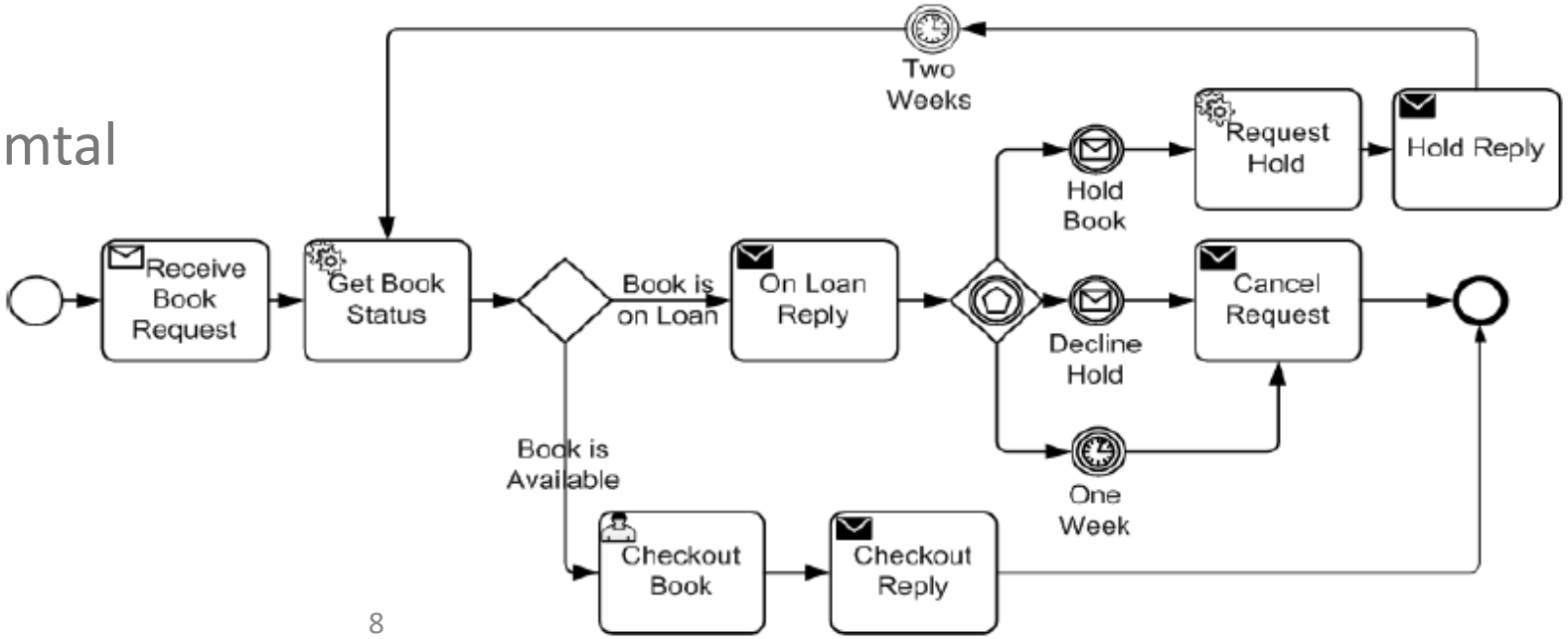
Verksamhetsarkitekten



- Arbetar med verksamhetens processer och informationsflödes i samverkan med IT-system
  - **Exempelvis:** Hur fungerar orderhanteringsprocessen? Finns utvecklingspotential i hur systemen används?
- Visualiserar verksamheten med **modeller** – nuläge och målbeskrivningar
- Ansvarig för att kraven på IT-stöd/system är kopplade till verksamhetens behov
- Hjälper organisationen att möta föränderliga omvärldskrav
- Kan vara en **domänarkitekt** (djup förståelse inom ett visst område)

Verksamhetsarkitektens output

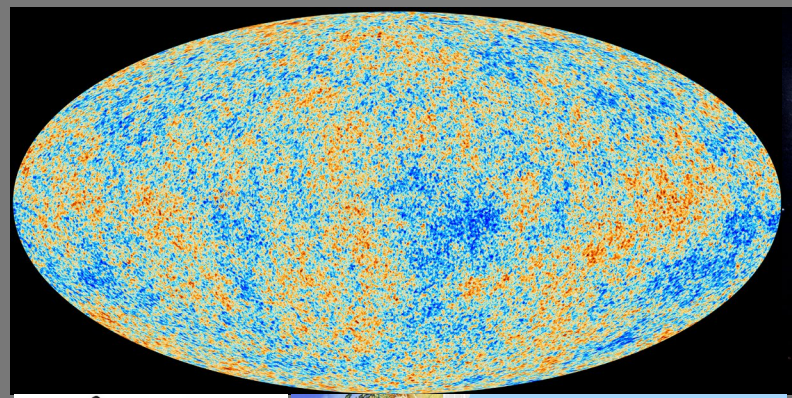
- Processbeskrivning (t.ex. BPMN)
  - **Exempel:** Hur hanteras en order?
  - **Exempel:** Hur lånas en bok ut?
- Insamlade klagomål
  - Intervjuer, personliga samtal
  - Helpdesk-ticket
  - Briefings

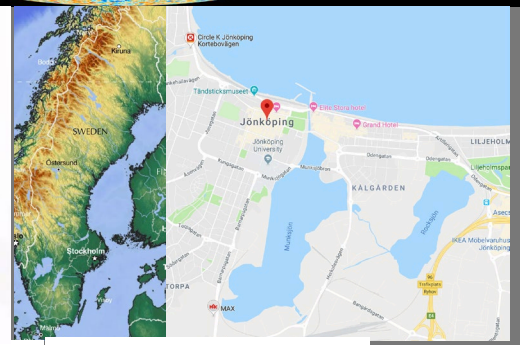


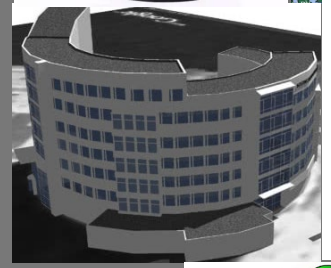
WHAT IS A MODEL?

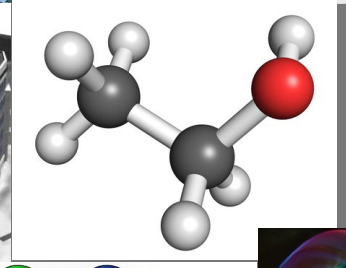
- **A model is a description of reality**
  - Models is a representation of reality
  - Usually a simplification
  - Helps to understand reality
- **All models are wrong, but some are useful!** 😉
- **The suitable complexity of models will depend on the purpose (e.g. understanding, prediction, evaluation, design…)**

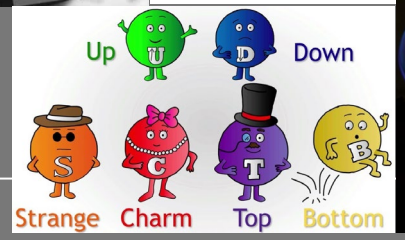
WHY DO WE NEED MODELS?

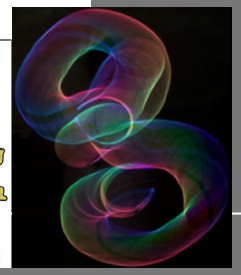












"No substantial part of the universe is so simple that it can be grasped and controlled without making abstractions where these abstractions represent the focused part of the universe." (Rosenblueth & Wiener, 1945)

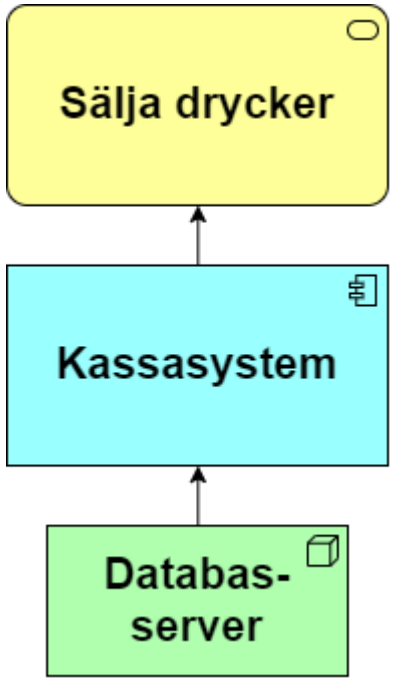
Ett inte helt ovanligt arkitekturscenario…

2011

Vad gör verksamheten?

Vilka system används?

Vilken teknik används?



2024

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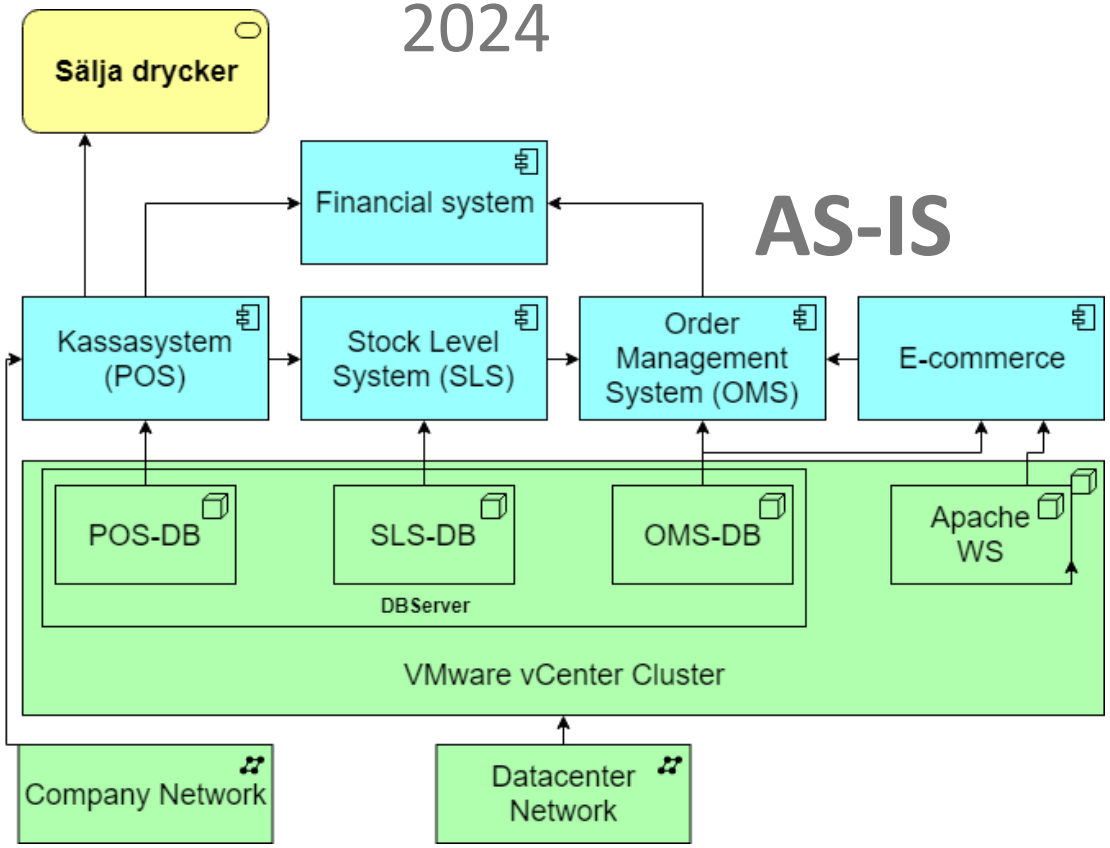
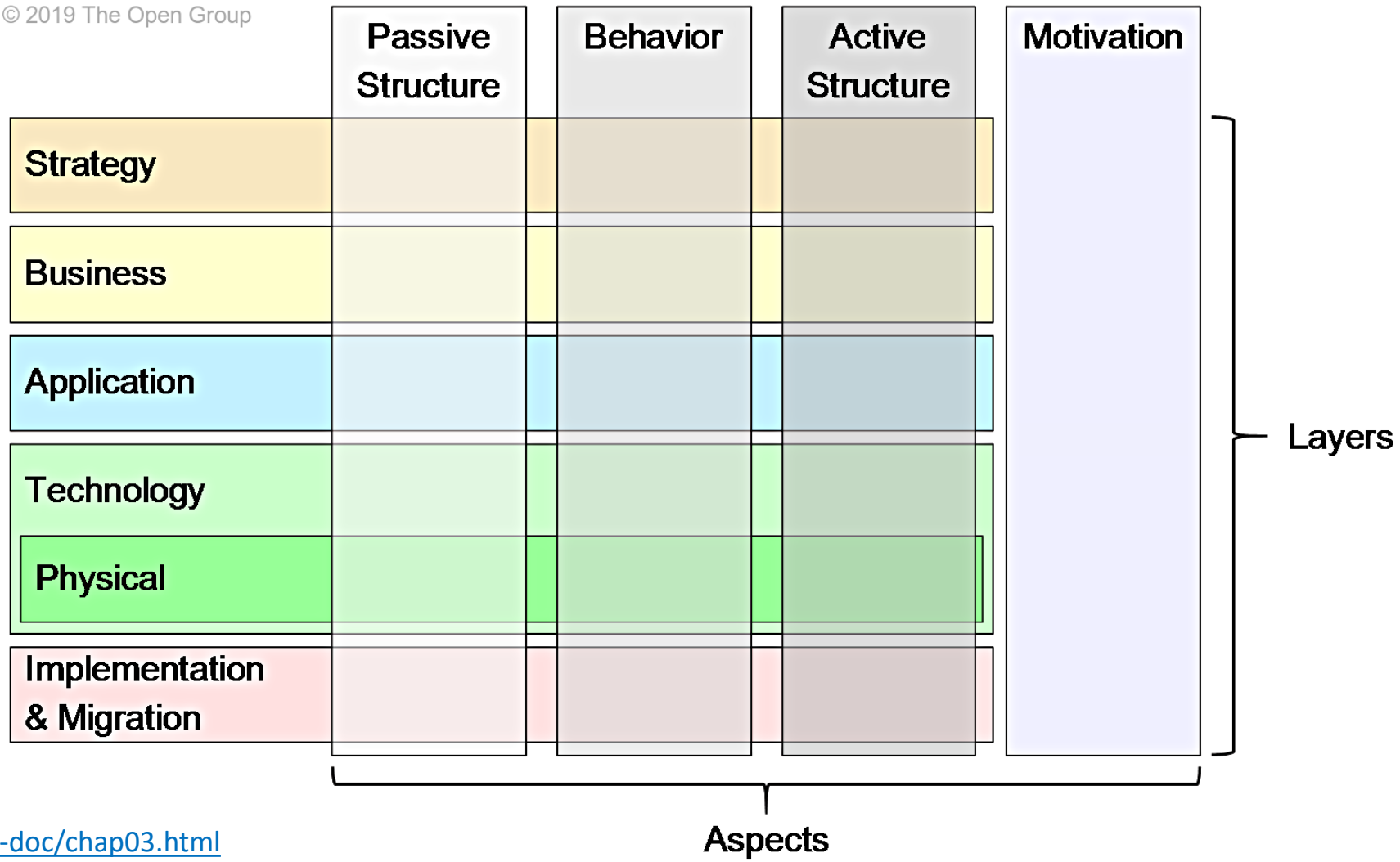


2030

TO-BE

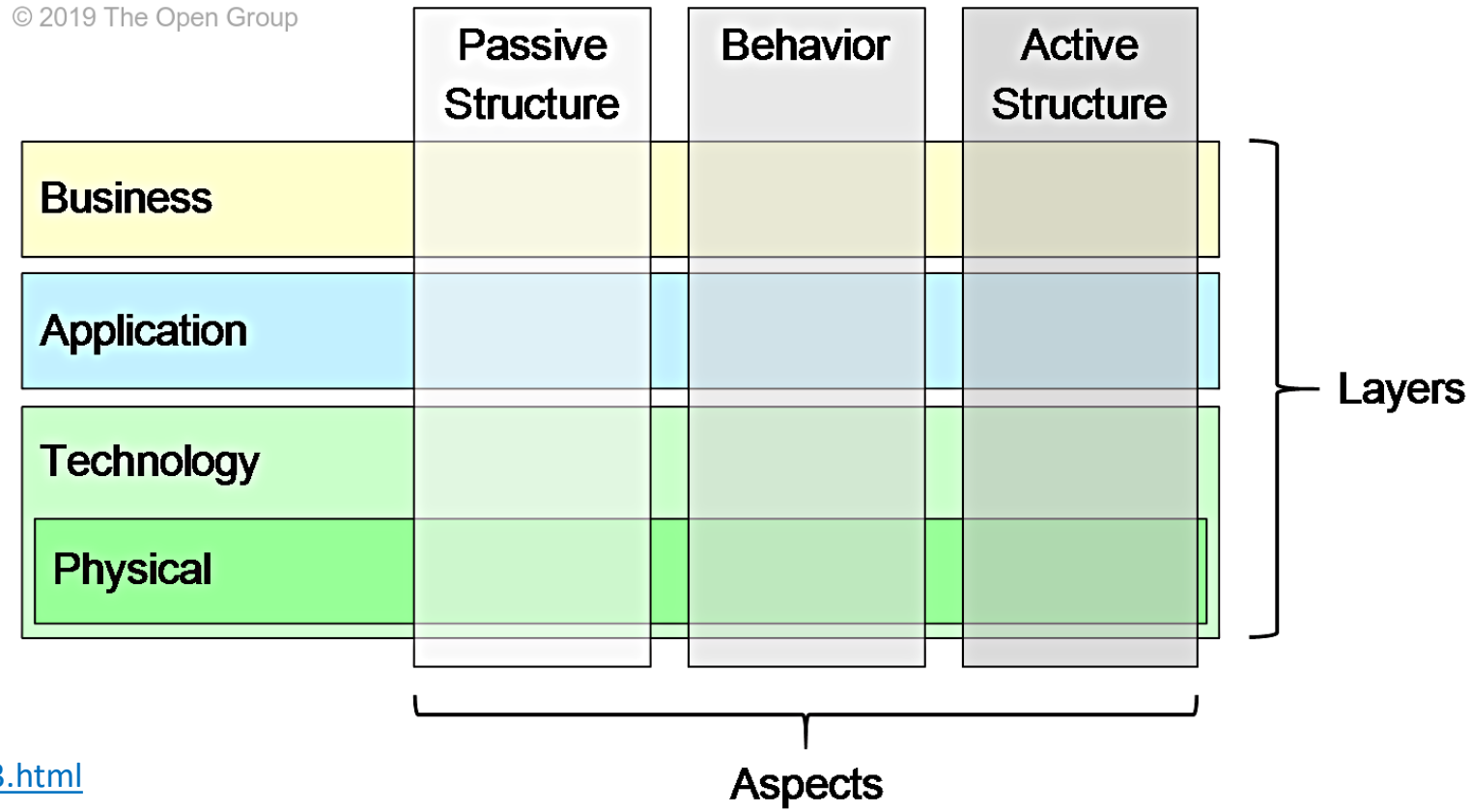
SYSTEM BOLAGET

ArchiMate Full Framework



https://pubs.opengroup.org/architecture/archimate3-doc/chap03.html

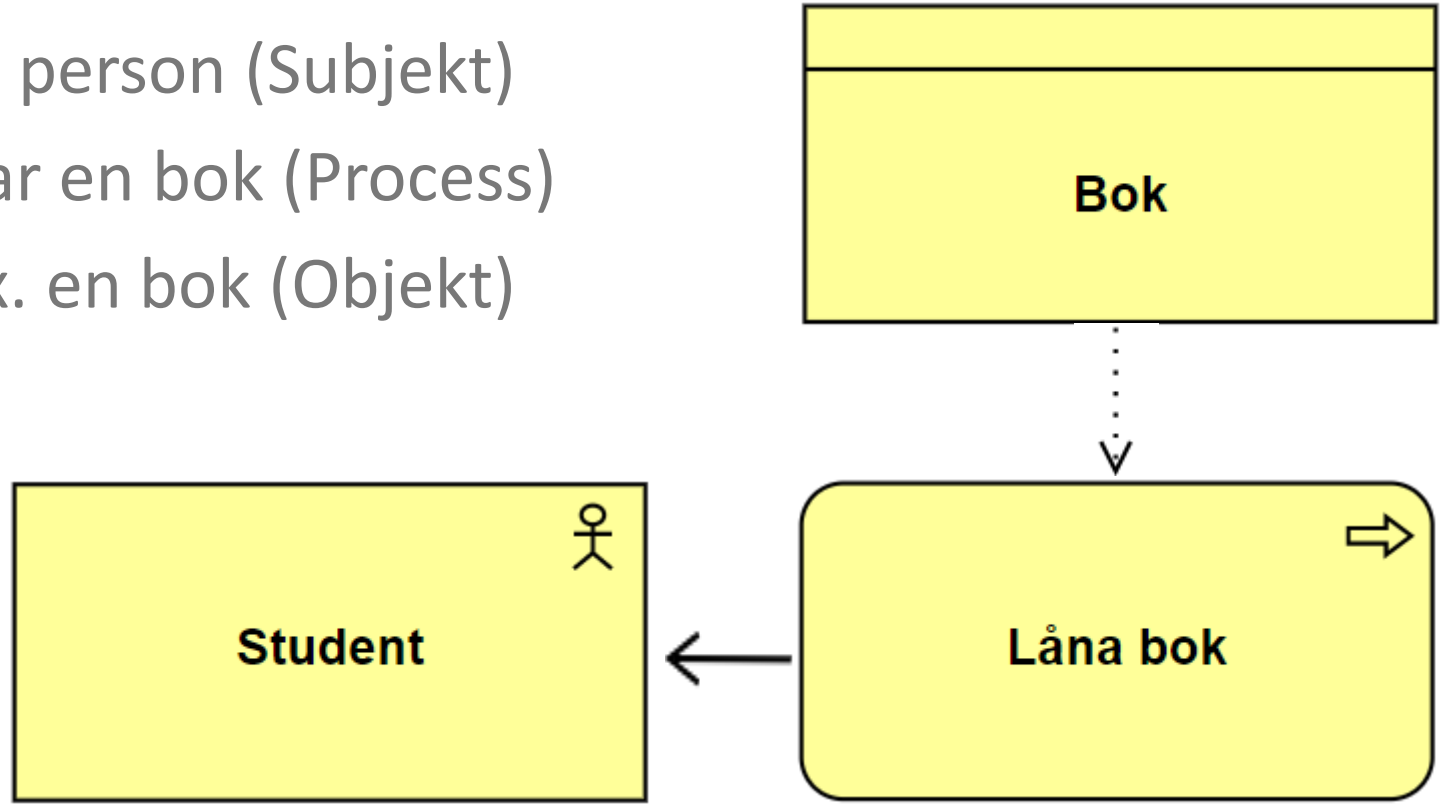
ArchiMate Core Framework



https://pubs.opengroup.org/architecture/archimate3-doc/chap03.html

Element

- **Aktiv struktur**, t.ex. person (Subjekt)
- **Beteende**, t.ex. lånar en bok (Process)
- **Passiv struktur**, t.ex. en bok (Objekt)



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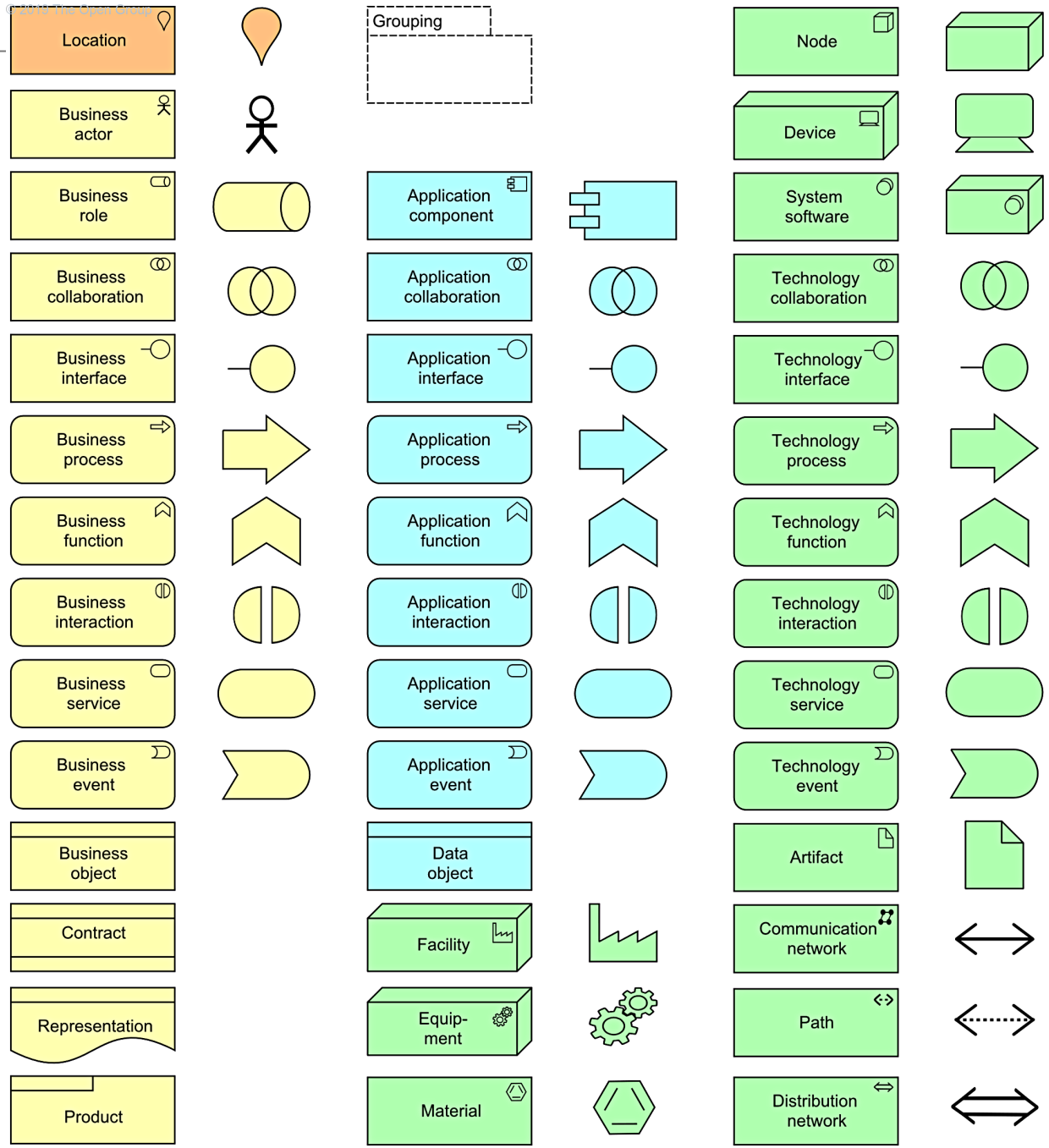
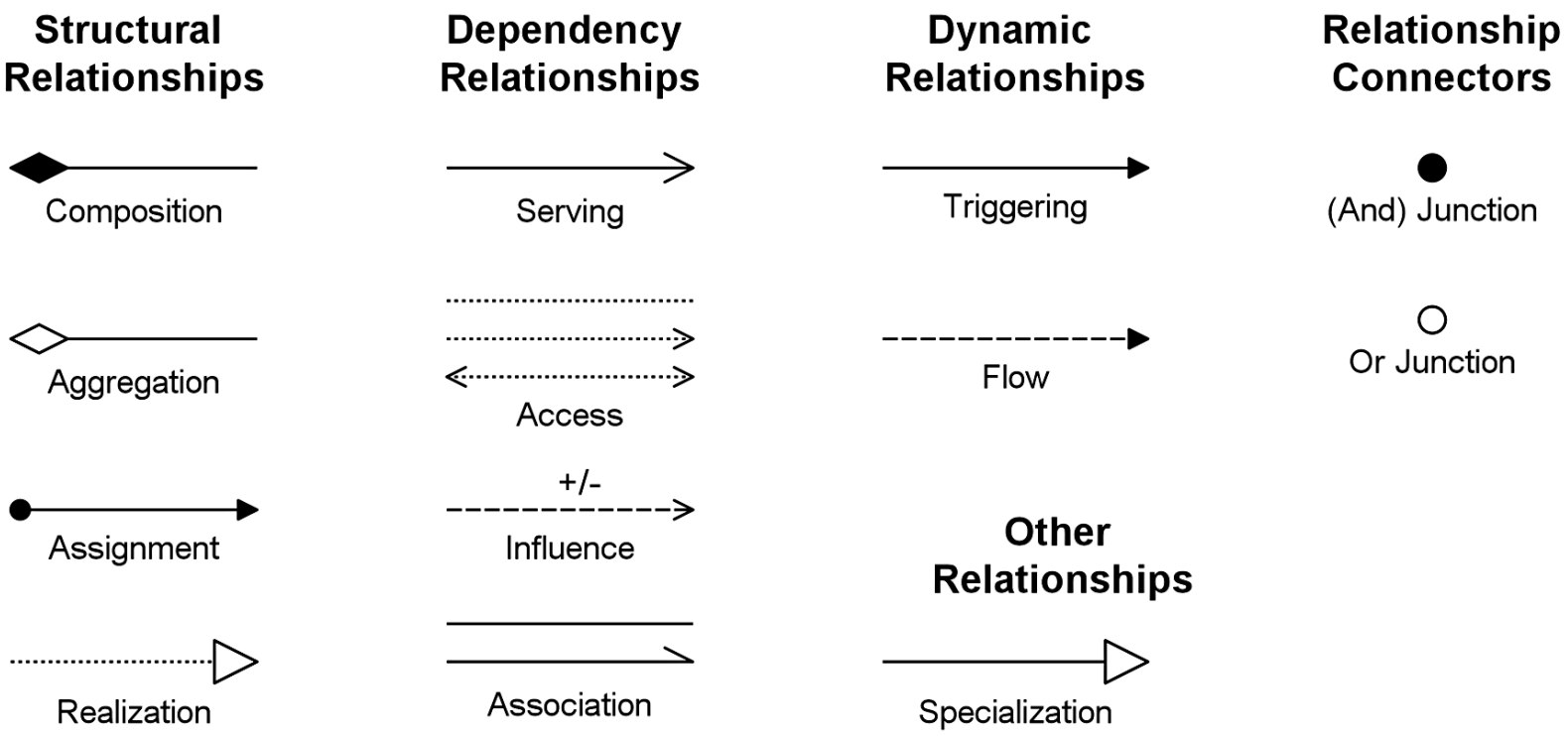
Element


Location
Business actor
Business role
Business collaboration
Business interface
Business process
Business function
Business interaction
Business service
Business event
Business object
Contract
Representation
Product
Grouping
Application component
Application collaboration
Application interface
Application process
Application function
Application interaction
Application service
Application event
Data object
Facility
Equip-ment
Material
Node
Device
System software
Technology collaboration
Technology interface
Technology process
Technology function
Technology interaction
Technology service
Technology event
Artifact
Communication network
Path
Distribution network

Relationer



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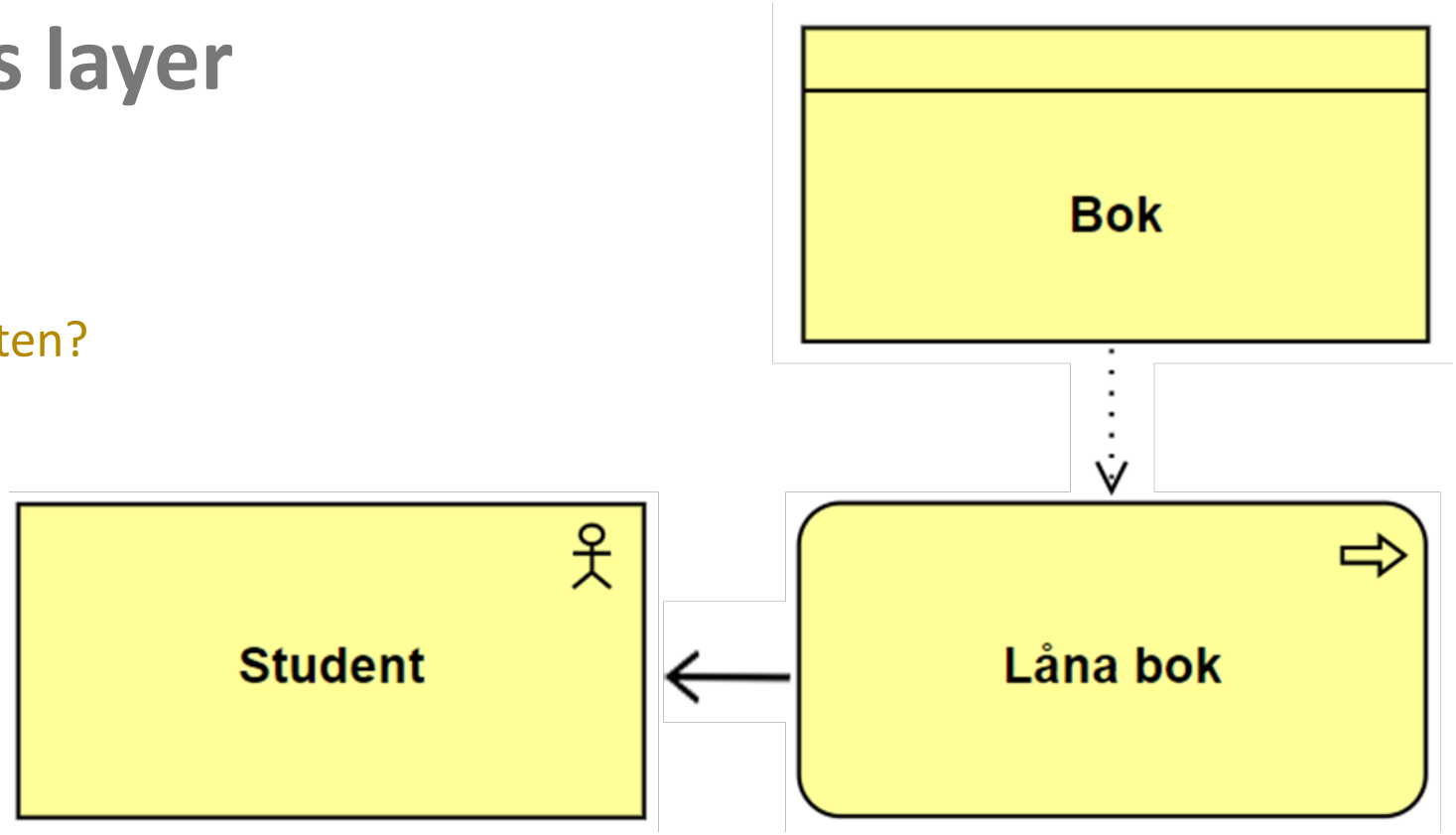
https://pubs.opengroup.org/architecture/archimate3-doc/chap05.html

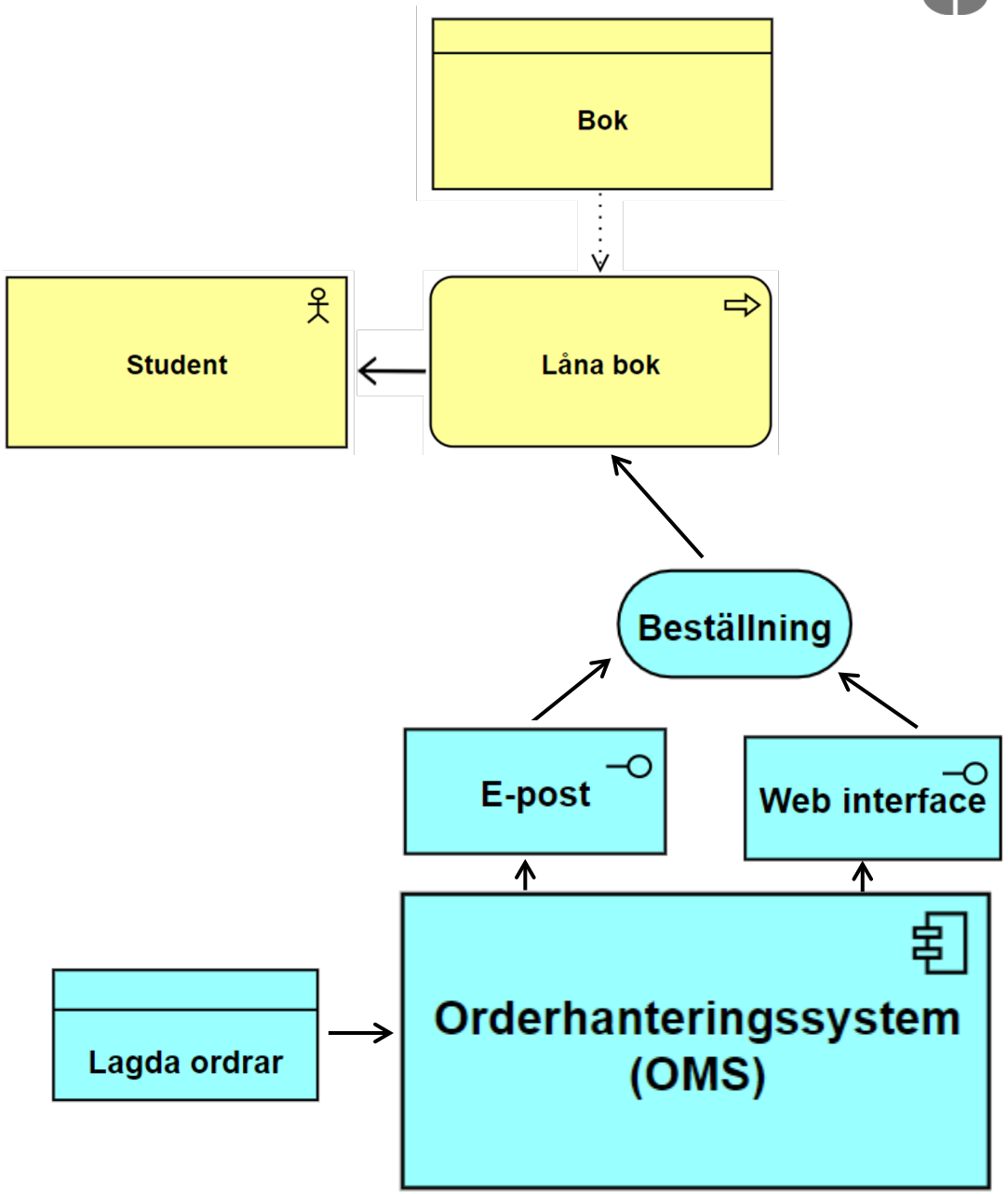
Business layer

Vad gör verksamheten?



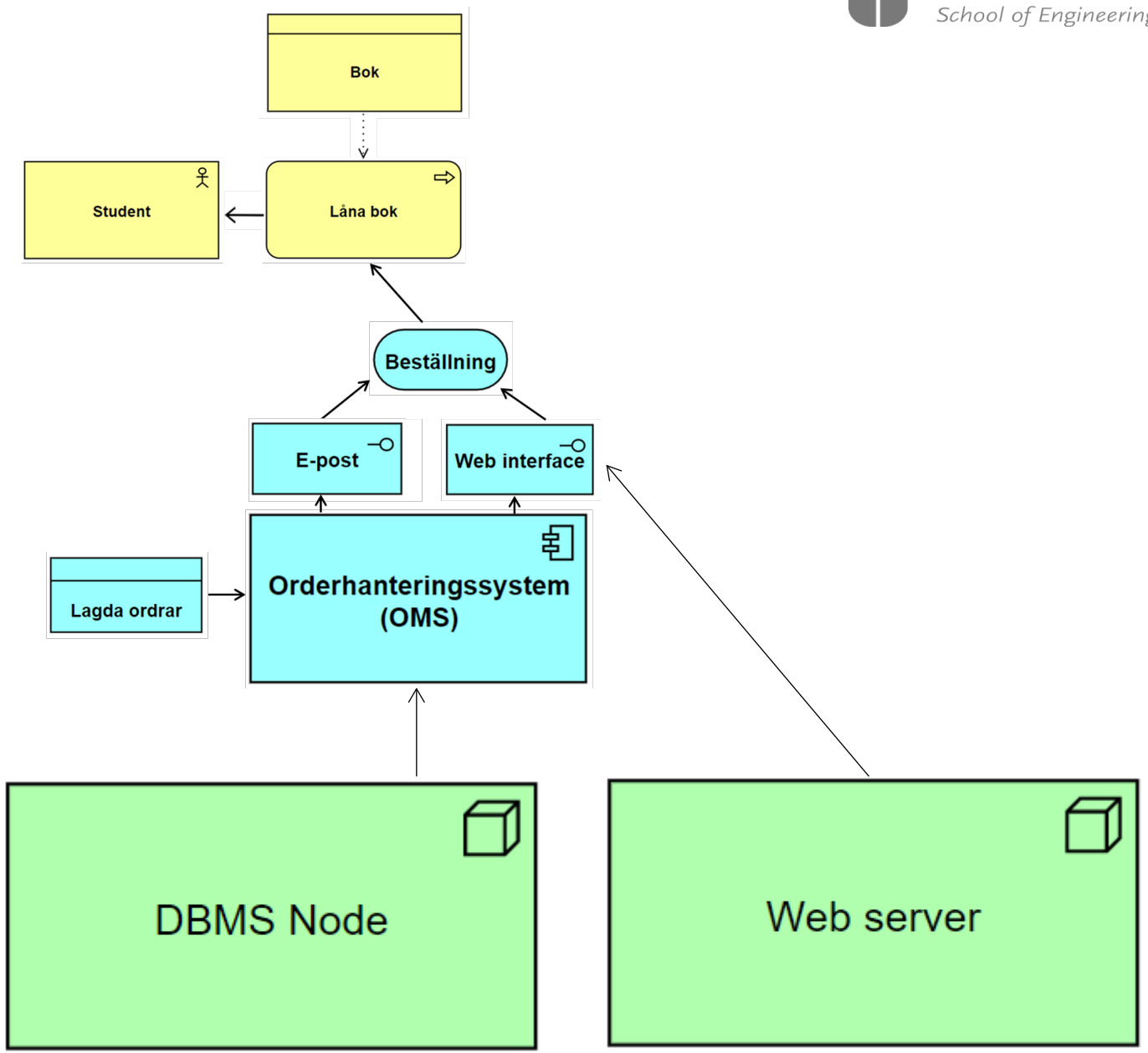
Application layer

Vilka system används?



Technology layer

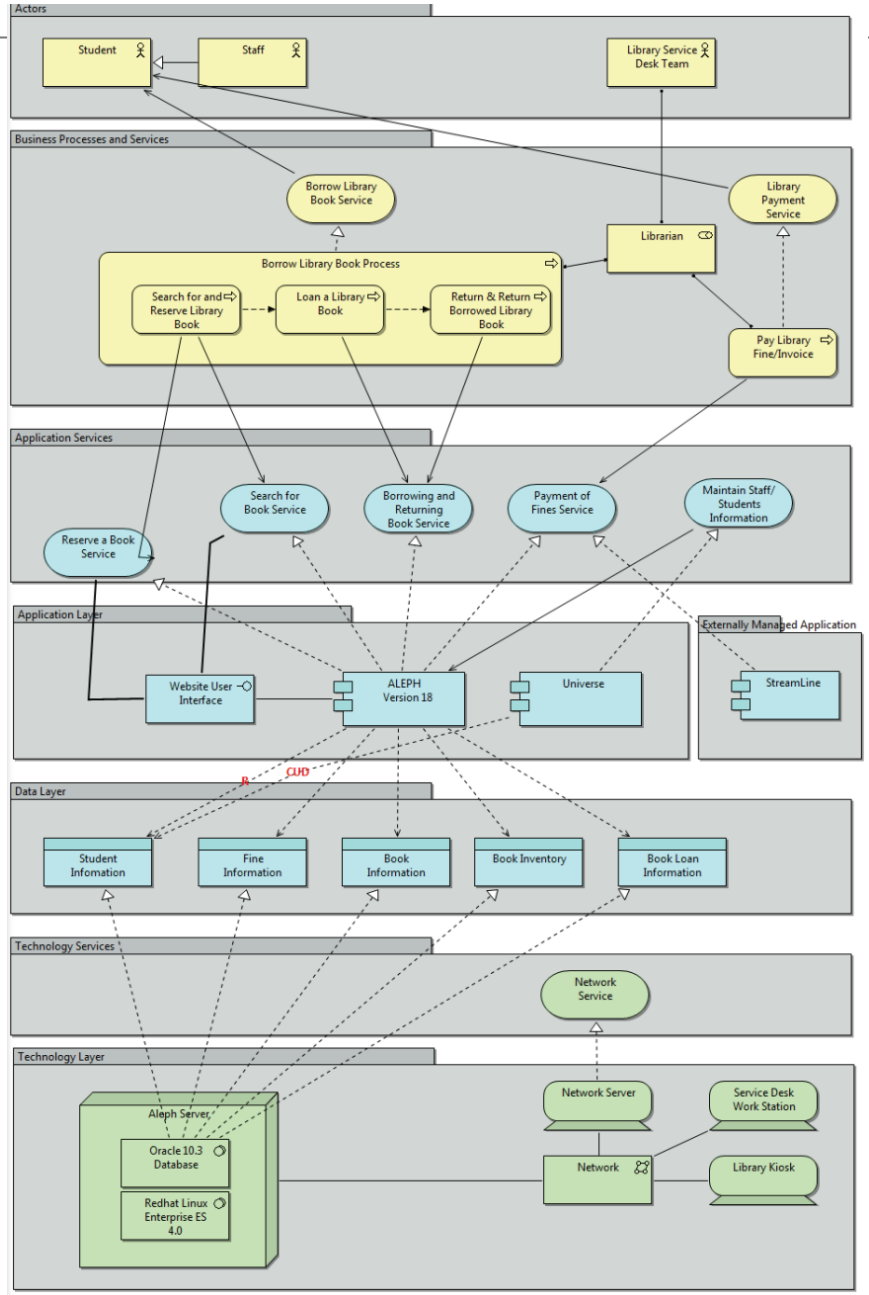
Vilken teknik används?



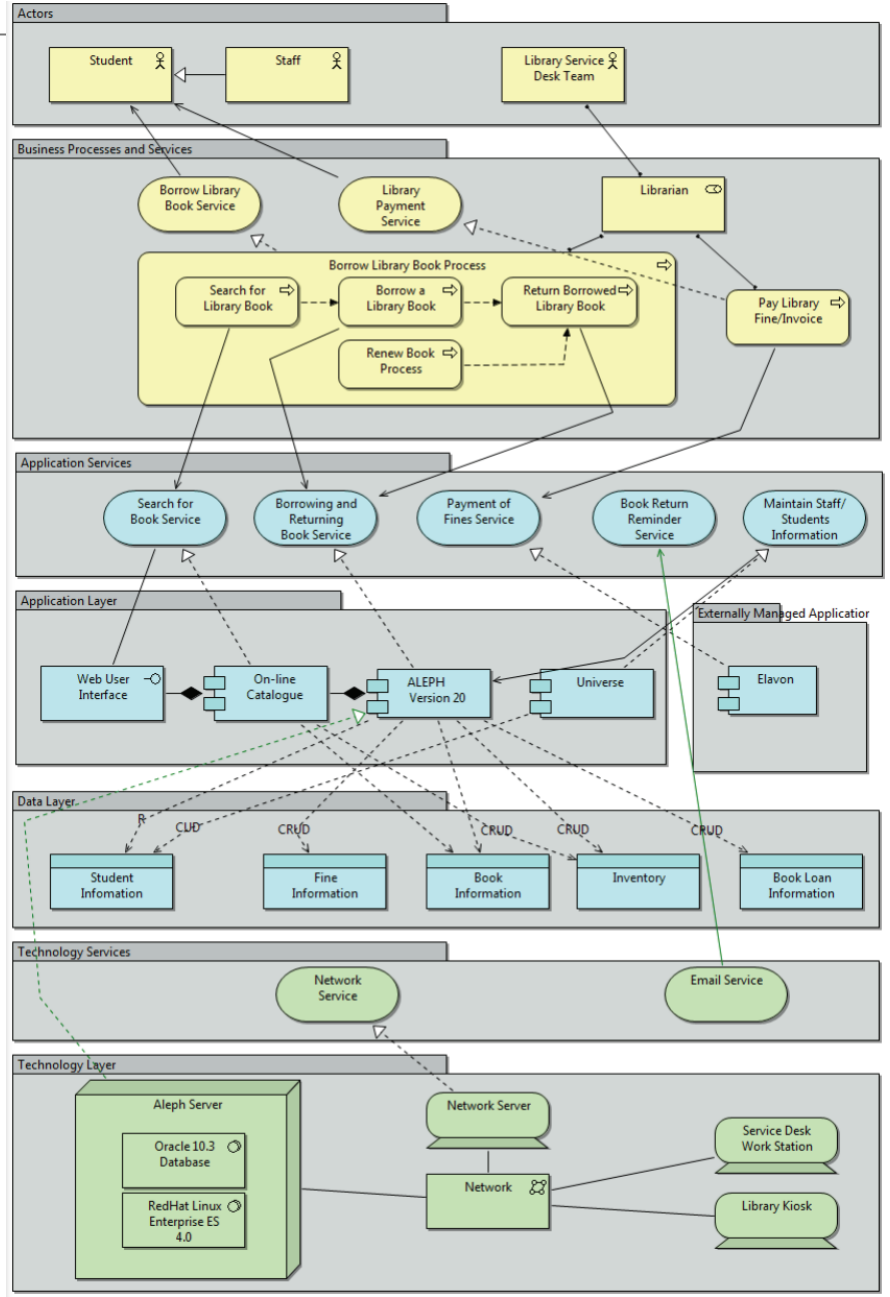
Ett bra exempel

Enterprise Architecture Evaluation Report, Conventry University

AS-IS


Actors
Student
Staff
Library Service Desk Team
Business Processes and Services
Borrow Library Book Service
Library Payment Service
Librarian
Borrow Library Book Process
Search for and Reserve Library Book
Loan a Library Book
Return & Return Borrowed Library Book
Pay Library Fine/Invoice
Application Services
Search for Book Service
Borrowing and Returning Book Service
Payment of Fines Service
Maintain Staff/ Students Information
Reserve a Book Service
Application Layer
Website User Interface
ALEPH Version 18
Universe
Externally Managed Application
StreamLine
R
CUD
Data Layer
Student Infomation
Fine Information
Book Information
Book Inventory
Book Loan Information
Technology Services
Network Service
Technology Layer
Aleph Server
Oracle 10.3 Database
Redhat Linux Enterprise ES 4.0
Network Server
Network
Service Desk Work Station
Library Kiosk

TO-BE


Actors
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Borrow Library Book Process
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Borrow a Library Book
Return Borrowed Library Book
Renew Book Process
Pay Library Fine/Invoice
Application Services
Search for Book Service
Borrowing and Returning Book Service
Payment of Fines Service
Book Return Reminder Service
Maintain Staff/ Students Information
Application Layer
Web User Interface
On-line Catalogue
ALEPH Version 20
Universe
Externally Managed Applicatior
Elavon
R
CUD
CRUD
CRUD
CRUD
CRUD
Data Layer
Student Infomation
Fine Information
Book Information
Inventory
Book Loan Information
Technology Services
Network Service
Email Service
Technology Layer
Aleph Server
Oracle 10.3 Database
RedHat Linux Enterprise ES 4.0
Network Server
Network
Service Desk Work Station
Library Kiosk

Uppgift 2 – Modellera i ArchiMate

- Läs igenom uppdragsbeskrivningen (Systembolaget)
- ArchiMate i draw.io:
  - Välj **Create New Diagram** -> **Basic** -> **Blank Diagram**
  - Gå sedan till **More Shapes** -> **Business** -> **ArchiMate 3.0**

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